

DUNITH DESITHA ATHUKORALA

FINAL YEAR UNDERGRADUATE

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PROFILE

Final-year Instrumentation & Automation Technology undergraduate, also pursuing a Bachelor of Information Technology at UCSC, with a strong foundation in reinforcement learning, computer vision, and full-stack software engineering. Applied deep learning and simulation to real-world control problems, and built full-stack applications using modern web and backend frameworks. Skilled in end-to-end AI and software system development, from data pipelines and model design through deployment.

EXPERIENCE

Automation Engineering Intern

Feb 2025 – Aug 2025

Phoenix Industries Ltd

- Designed and fabricated a centrifugal bottle feeder (bottle unscrambler) in SolidWorks for 25 mL bottles, reaching a tested throughput of 16 bottles/minute
- Developed a 10-node distributed audio system using Raspberry Pi 4 Model B and ESP8266 boards, with MQTT for wireless device communication and control
- Assisted in a YOLO-based computer vision system (Roboflow, Google Colab) for defect detection and quality classification
- Created and modified 3D CAD models and mechanical assemblies in SolidWorks and supported fabrication and assembly
- Assisted with troubleshooting, maintenance, and testing of packaging and manufacturing equipment
- Prepared project plans, cost estimates, and component-tracking documentation to support engineering projects

EDUCATION

Bachelor of Engineering Technology (Hons) in Instrumentation and Automation

Jul 2022 – Aug 2026

Faculty of Technology, University of Colombo

- Relevant coursework: Digital Signal Processing, Sensors & Transducers, Data Acquisition Systems, Introduction to Robotics, Introduction to FPGA, Electronic Circuit Simulations, Technical Drawing & CAD, Industrial Automation & Control, Statistical Data Analysis, Data Communication & Networking
- Final exams and research are fully completed; awaiting results (Expecting graduation 2027)

Bachelor of Information Technology

Sep 2024 – Present

University of Colombo School of Computing (UCSC)

- Relevant coursework: Programming, Data Structures & Algorithms, Database Systems, Web Application Development, Object-Oriented Analysis & Design, Computer Networks, Agile Software Development, Mathematics for Computing

GCE Advanced Level (Engineering Technology Stream)

Aug 2018 – Oct 2020

Mahanama College, Colombo 03

- Engineering Technology: B | Science for Technology: A | ICT: B | General English: B

PROJECTS

AI-Based Adaptive Traffic Signal Optimization under Irregular Traffic Conditions

Oct 2025 – Jul 2026

Final Year Research Project (Completed)

- Designed and deployed an end-to-end AI system combining deep reinforcement learning (Dueling Double DQN with SE-attention) and YOLO-based computer vision to outperform fixed-time traffic controllers
- Integrated the trained model with ESP32 embedded hardware, taking the system from SUMO simulation through to real-time physical deployment

Weather Analytics Application – Take-Home Assignment

Aug 2026

- Built a full-stack app (Node.js, Express, TypeScript backend; React, TypeScript, Vite frontend) with Auth0 authentication (JWT, MFA) and a self-designed Comfort Index scoring algorithm based on Thom's Discomfort Index and ASHRAE 55
- Implemented in-memory caching, unit tests (Jest), and Docker Compose containerization; documented trade-offs and design decisions

EV Price Predictor – Self-Learning Project (Ongoing)

Mar 2026 – Present

- Constructed an ML pipeline on a 3,022-instance dataset, benchmarking 3 models (Linear Regression, Random Forest, XGBoost) with MLflow experiment tracking
- Architected a FastAPI + PostgreSQL inference service with a React/TypeScript frontend, containerized via Docker Compose with a GitHub Actions CI/CD pipeline

RAG System – Self-Learning Project

Mar 2026 – Present

- Created a RAG document Q&A system (Gemini, LangChain, Chroma) with OCR ingestion and citation-grounded responses, deployed via Streamlit

Smart Home Garden DAQ System (GreenHeaven) – DAQ Mini Project

Oct 2024 – Nov 2024

- Built an ESP32-based IoT environmental monitoring system with a real-time sensor-to-dashboard pipeline, presented as a live demo showcasing threshold-based automation design for smart irrigation

Customer Satisfaction of Tiles in Sri Lanka – Statistical Data Analysis Coursework Project

Mar 2024 – Jun 2024

- Led statistical analysis of 200 survey responses (SPSS: T-Test, ANOVA, Regression) across 25 districts, identifying promotions as the top satisfaction driver

SKILLS

- Programming:** Python, C++, Java, JavaScript, SQL
- AI / Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, CNN, LSTM, YOLOv11, YOLOv26, Dueling DQN, Prioritized Experience Replay, Squeeze-and-Excitation (SE) Attention
- Computer Vision & Data:** OpenCV, Roboflow, Data Annotation/Labelling, Multi-camera Systems, NumPy, Pandas, Matplotlib
- Statistical Analysis:** SPSS, Descriptive Statistics, Hypothesis Testing (t-Test, ANOVA), Multiple Linear Regression, Reliability Analysis
- LLM & Generative AI:** LangChain, Hugging Face, RAG, Chroma, Vector Databases, Graph Databases, Graph RAG, Hybrid Search, Prompt Engineering, Ollama, OpenAI API, Google Gemini API, Anthropic Claude API
- Web & Backend:** Flask, FastAPI, Node.js, Express.js, React, REST APIs, Streamlit, SQLAlchemy, PostgreSQL, MongoDB, SQLite, HTML, CSS, XML
- CS Fundamentals:** OOP, Data Structures & Algorithms, DBMS, Computer Networking, Agile Software Development, Enterprise Application Development
- Embedded Systems & IoT:** Arduino, Raspberry Pi, ESP32/ESP8266, Microcontroller Programming
- Tools & Platforms:** Git, GitHub, Docker, MATLAB/Octave, Google Colab, Kaggle, Arduino IDE
- CAD & Fabrication:** SolidWorks, 3D Printing, Basic CNC Fabrication

LEADERSHIP ROLES / VOLUNTEERING

- SEDS UOC** – Assistant Secretary (2024/25)
- Colombo Beacon** – Top Board Director (2024/25) Middle Board Director (2023/24)
- Techno Media** – Head of the Content Writing Department (2024/25)

REFERENCES

Dr. Hansika Atapattu Senior Lecturer, Department of Instrumentation & Automation Technology, Faculty of Technology, University of Colombo Email: hansika@iat.cmb.ac.lk Contact: +94 717 669 711	Mr. Lahiru Randika Ekanayake Assistant Manager, Automation, Phoenix Industries Limited, Industrial Zone (East), Makandura, Gonawila Email: lahiru@phoenix.lk Contact: +94 715 854 456
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